

# Historical Global Tropical Cyclone Intensity Trends

Aniket Dev Roy<sup>1</sup>, Aaron Kruskie<sup>2</sup>, and Dan R. Chavas<sup>3</sup>

<sup>1</sup>Department of Earth, Atmospheric, and Planetary Sciences, Purdue University, West Lafayette, IN, USA

## Key Points:

- Why do we observe differences in tropical cyclone intensities when comparing datasets that use different observational techniques?
- How do these changes vary across all wind intensities versus only a certain group?
- How does the mean wind intensity of a tropical cyclone shift in relation to maximum wind intensity?

## Abstract

Tropical cyclones (TCs) are among the most destructive weather systems globally, with increasing evidence suggesting that their intensity has risen in recent decades under a warming climate. This study investigates global and basin-scale trends in TC intensity using the International Best Track Archive for Climate Stewardship (IBTrACS) dataset. Key parameters, including maximum sustained wind speed ( $V_{\max}$ ), minimum sea-level pressure (MSLP), and potential intensity (PI), were analyzed on a  $4^\circ \times 4^\circ$  grid. Linear regression and Theil–Sen estimators were applied to assess temporal trends in these variables and their ratios. The results are compared with findings from previous studies, including (Kossin et al., 2020; Kruskie & Chavas, n.d.), to evaluate consistency and identify regional differences in intensification. Our analysis aims to clarify whether the observed global increase in TC strength is uniform across all storm categories or primarily driven by a subset of high-intensity systems.

## 1 Introduction

Tropical cyclones are considered one of the most dangerous and economically damaging weather phenomena in the world. These “super” storms cause widespread disturbances in human habitations such as the destruction of infrastructure, agriculture, and lives. The motivation to understand them has gained a lot of momentum in the past decade, all thanks to recent studies that have shown an increase in the average intensity of all tropical cyclones ranging from weak to strong systems that occur globally. This positive shift in intensity has been attributed to changes in climate, i.e. global warming, making them more destructive in nature. Hence, newly found knowledge of them becomes critical at this time in order to understand their behavioral patterns in order to minimize the damages they could potentially cause in the future. This study aims to add to the already vast knowledge of ongoing tropical cyclone research by understanding this “shift” in intensity levels by comparing and analyzing datasets that use different observational techniques to capture features of tropical cyclone systems, such as wind speed, pressure, potential intensity, and comparing them with the results of widely accepted published research. Based on the study by (Kossin et al., 2020), which used homogeneous tropical cyclones around the world from 1979-2017, showed that there is an increase of about 8 percent per decade in the exceedance probability (a measure of crossing a threshold in the future) of major tropical cyclones (Saffir-Simpson Categories 3-5) with a 95 percent confidence interval of 2-15 percent per decade. This is consistent with the recent climate change trend. These trends are seen to be more pronounced in the North Atlantic (49 percent) and southern Indian Ocean (18 percent) with modest to insignificant changes in other basins such as the Pacific. In summary, recent studies have shown that all systems ranging from weak to strong, i.e. category 1 to 5 of tropical cyclones, have experienced an intensification of their strengths. When looking at the analysis conducted by us, we questioned whether in reality all tropical cyclone systems have undergone this change or have a certain group of them experienced this increased strength. Our study works with the IBTrACS dataset to better understand this subject.

## 2 Methods

### 2.1 Data Sources

This study utilized two primary datasets for analyzing tropical cyclone (TC) characteristics: the International Best Track Archive for Climate Stewardship (IBTrACS) and the Hurricane Satellite (HURSAT) dataset. IBTrACS is a comprehensive global repository of TC best-track data, aggregating observations from multiple meteorological agencies worldwide. It provides standardized records of TC parameters, including position (latitude and longitude), maximum sustained wind speed ( $V_{\max}$ ), minimum sea-level

pressure (MSLP), and track information, spanning from 1842 to the present with temporal resolutions typically at 3- or 6-hour intervals. The HURSAT dataset complements IBTrACS by providing homogenized satellite-derived observations of TCs. Derived from the International Satellite Cloud Climatology Project (ISCCP) B1 infrared imagery and other satellite sources, HURSAT offers TC-centric gridded data in netCDF format, including brightness temperatures and intensity estimates, for global basins from 1980 onward. (the only downside to this is that it is currently available only till 2016). Potential Intensity (PI) values, used for computing the  $V_{\max}/PI$  ratio, were derived from thermodynamic principles based on sea surface temperature (SST) and atmospheric profiles. PI represents the theoretical upper limit of TC wind speed.

## 2.2 Data Processing and Gridding

Tropical cyclone (TC) data were processed on a global scale, focusing on the major basins: North Atlantic (NA), Western North Pacific (WNP), Central North Pacific (CNP), Eastern North Pacific (ENP), Southern Indian Ocean (SI), and South Pacific (SP). TC tracks were filtered to include only observations where the maximum sustained wind speed ( $V_{\max}$ ) exceeded 64 knots (hurricane intensity) and later where  $V_{\max} > 34$  knots (tropical storm intensity).

Spatial aggregation was performed on a  $4^\circ \times 4^\circ$  latitude-longitude grid to compute mean values and spatial densities. For each grid cell, the following variables were derived:

- **Track Density:** Number of unique TC tracks passing through the cell, normalized per year.
- **Wind Intensity:** Mean  $V_{\max}$  ( $\text{m s}^{-1}$ ) across all TC observations within the cell.
- **MSLP:** Mean minimum sea-level pressure (mb) for TCs within the cell.
- **$V_{\max}/PI$  Ratio:** Ratio of observed  $V_{\max}$  to the theoretically computed potential intensity (PI), averaged per grid cell.

Data were interpolated onto the grid using a nearest-neighbor assignment for point observations to preserve observed intensity distributions and avoid artificial smoothing.

## 2.3 Trend Estimation

Trends in each variable were estimated using **Theil-Sen Estimator**, a robust method to estimate linear trend devoid of outliers. This approach is less sensitive to outliers and non-normal distributions, making it suitable for TC data that may contain extreme events or observational artifacts. The estimator was computed using the `scipy.stats.theilslopes` function.

## 2.4 Study Hypotheses

Recent work has documented increases in the probability of major tropical cyclones (TCs) in a warming climate (Kossin et al., 2020). Motivated by these findings and by the global and basin-scale heterogeneity evident in prior studies, this analysis is designed to test the following hypotheses:

1. **Global intensification:** Tropical cyclone intensity has increased over the improvement in observational techniques, such that long-term linear trends in maximum sustained wind speed ( $V_{\max}$ ) are positive in at least some basins during 1990–2024.
2. **Dependence on storm strength:** Intensification signals are stronger and more spatially coherent for major hurricanes ( $V_{\max} \geq 64$  kt) than for the broader population of weaker systems ( $V_{\max} \geq 34$  kt).

3. **Dataset dependence:** Trend structures inferred from IBTrACS and HURSAT are qualitatively similar at large scales but differ in magnitude, with IBTrACS exhibiting larger-amplitude trends due to heterogeneous operational reporting, and HURSAT providing more muted, homogenized signals.
4. **Periodic Shift in Intensity:** Intensification signals are not temporally constant; trends computed over more recent decades (e.g., 1990–2010) show stronger positive tendencies than those over earlier intervals.

The subsequent results are interpreted in the context of these hypotheses, with particular emphasis on distinguishing whether the observed positive shifts in intensity are driven by the full TC population or are concentrated within a subset of high-intensity storms.

### 3 Results

#### 3.1 Track Density

The mean track density of strong tropical cyclones (hurricanes) from 1990–2024 shows distinct spatial patterns across global basins. The Western North Pacific (WNP) exhibits the highest densities, particularly over the Philippine Sea, eastern China, and Japan, where values exceed 8–10 tracks per  $4^\circ \times 4^\circ$  grid cell. This indicates that the WNP remains the most active region for tropical cyclone formation and propagation. Moderate to high densities are also evident in the North Atlantic, concentrated over the Caribbean Sea, Gulf of Mexico, and along the U.S. East Coast, where peak values range between 6–8 tracks per grid cell.

Trends in track density reveal spatially heterogeneous changes across basins. A positive trend of approximately  $+0.18$  tracks  $\text{yr}^{-1}$  is observed over the WNP, suggesting a gradual increase in the frequency of strong hurricanes in that region. However, adjacent areas show localized decreases of about  $-0.27$  tracks  $\text{yr}^{-1}$ . The North Atlantic exhibits relatively subtle variations, not too significant. A pronounced negative trend is evident along the coast of California ( $-0.18$  tracks  $\text{yr}^{-1}$ ), indicating potential decrease in storm occurrence there. The Theil–Sen estimator confirms statistically significant trends primarily in the WNP and along the coast of California.

**Research Question:** Is the frequency of strong tropical cyclones increasing globally, or are these increases region specific?

These track density trends as seen in WNP basin and along the coast of California indicate that increases in global frequency of strong system hurricanes is not uniform but region specific.

#### 3.2 Wind Intensity

Wind intensity trends for strong systems, evaluated using Theil–Sen estimator exhibited spatially heterogeneous patterns across the global tropics and subtropics. The results show notable regions of strengthening storms, with positive trends reaching up to approximately  $1.05 \text{ m s}^{-1} \text{ yr}^{-1}$  in portions of the western North Pacific, and in the central North Atlantic. These areas indicate a gradual but persistent intensification of tropical cyclone winds over the study period.

Conversely, significant negative trends—as low as about  $-0.70 \text{ m s}^{-1} \text{ yr}^{-1}$ —are observed across the ENP, South Atlantic, specifically in the Gulf of America, and portions of the South Indian, around the region east of Madagascar. Additionally, pronounced clusters of strong positive trends appear throughout the North Atlantic, indicating a broader regional tendency toward increasing tropical cyclone wind intensities. Globally, overall, a lot more positive intensification is observed.



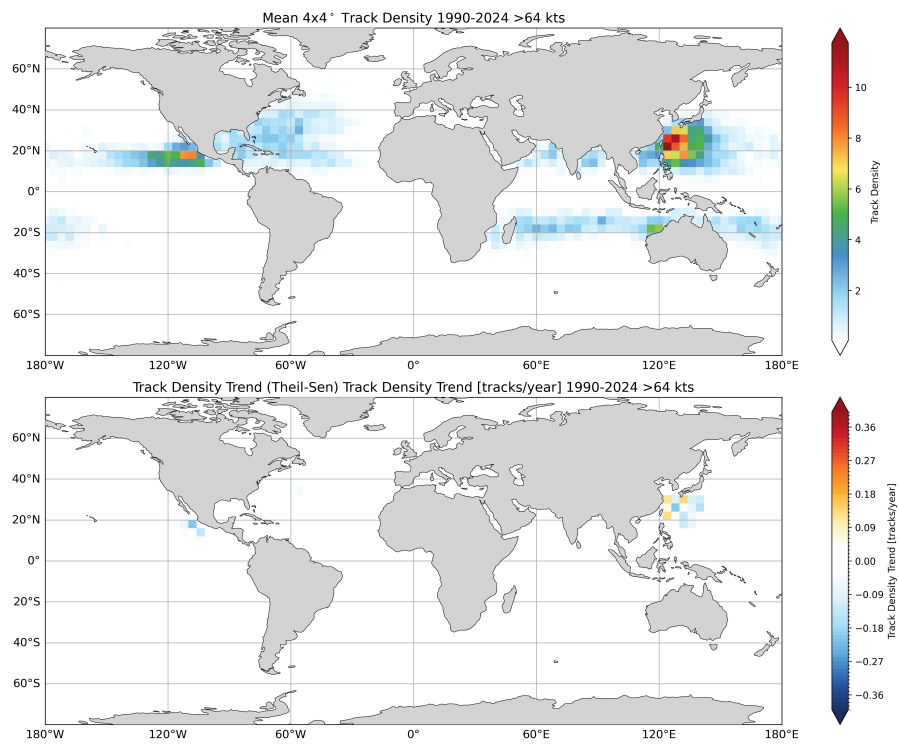


Figure 1: Mean track density of strong tropical cyclones from 1990–2024 and associated linear regression trends. Warm colors indicate increasing storm frequency, while cool colors represent decreases measured in tracks per year.

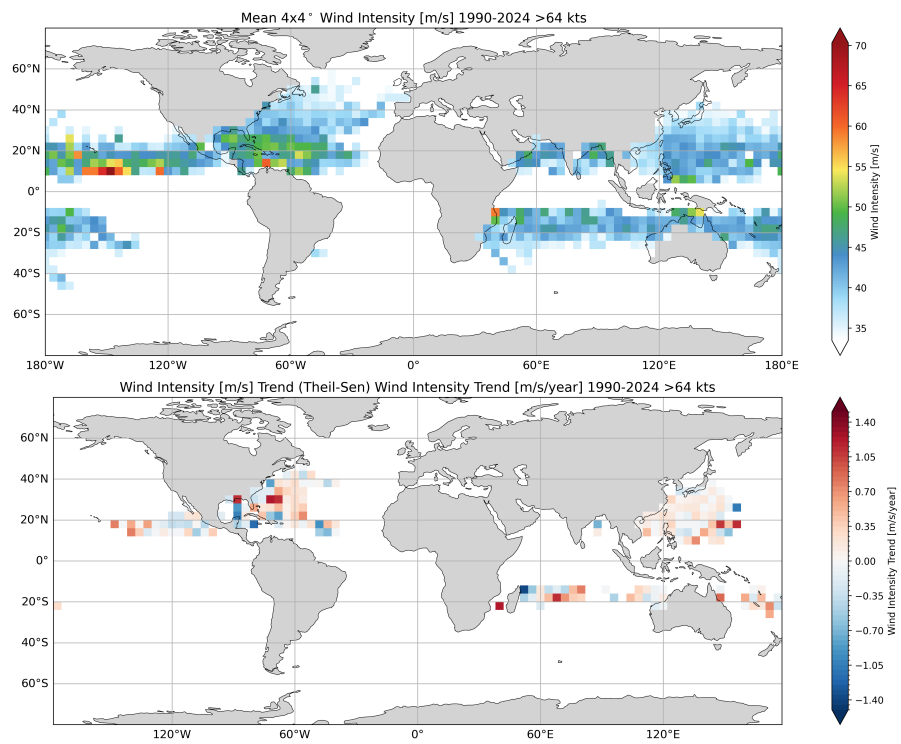


Figure 2: Spatial distribution of mean maximum sustained wind speed (Vmax) and Theil-Sen trends for strong storms, 1990–2024. Positive trends indicate intensifying storms, while negative trends show weakening.

The strengthening in the NA and with weakening in the SA, ENP, SP suggests intensification is not globally homogeneous. It also possible to have heterogeneous changes in this shift in intensification which can be either positive or negative within the same basin.

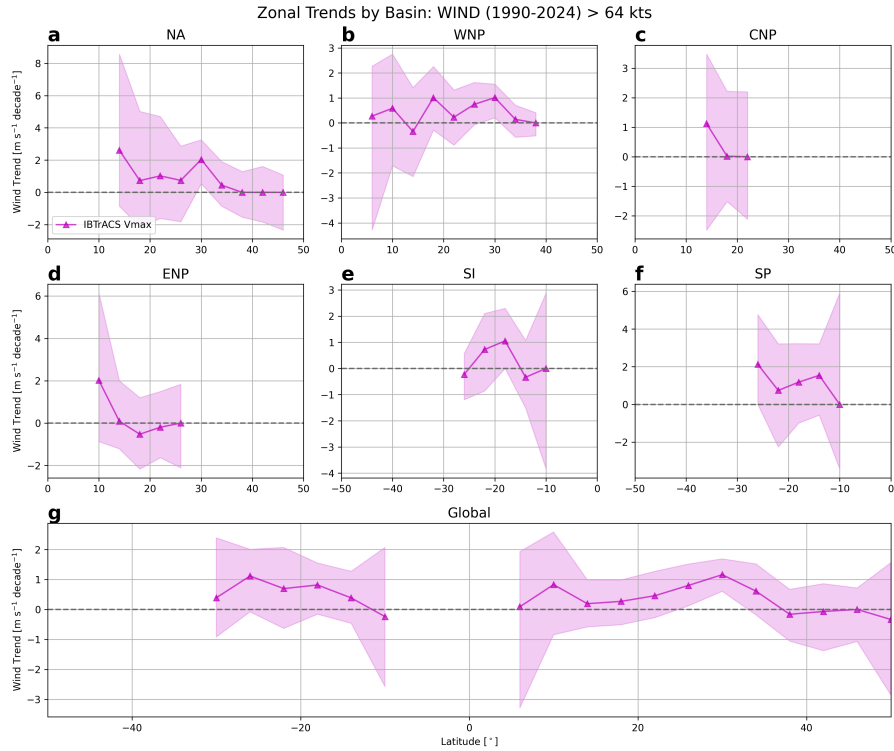


Figure 3: Basin specific wind speed (Vmax) and its Theil–Sen trends for strong storms, 1990–2024. NA undergoes a greater intensification relative to other basins.

The preferential intensification of major hurricanes in the NA relative to the ENP and WNP (Figure 3) is qualitatively consistent with the combination of long-term Atlantic warming and repeated La Niña events over the study period, which tend to reduce vertical wind shear and enhance potential intensity over the tropical Atlantic while suppressing activity in parts of the eastern Pacific.

### 3.3 Vmax/PI Ratio

The mean ratio of maximum wind speed (Vmax) to potential intensity (PI) over 1990–2024, restricted to wind speeds of at least 64 knots, provides a measure of how closely hurricanes approach their theoretical upper limits. High ratios (0.8–1.0) are widespread in the Subtropics as these regions efficiently utilize their environmental potential. Lower ratios (0.4–0.6) appear in the Tropics, where storms frequently fall short of PI due to factors such as increased vertical wind shear or entrainment of dry air.

Trends in the Vmax/PI ratio reveal only subtle changes rather than strong basin-wide signals. Results indicate positive trends of up to approximately  $+0.01 \text{ yr}^{-1}$  in parts of the south of western North Pacific and South Indian basins, suggesting that storms in these regions are potentially gradually approaching a larger fraction of their theoretical maximum intensity. Negative trends, reaching about  $-0.01 \text{ yr}^{-1}$ , are found in peripheral regions such as the eastern North Pacific, and near Madagascar.

**Research Question:** Does max wind intensity shift with potential intensity ?

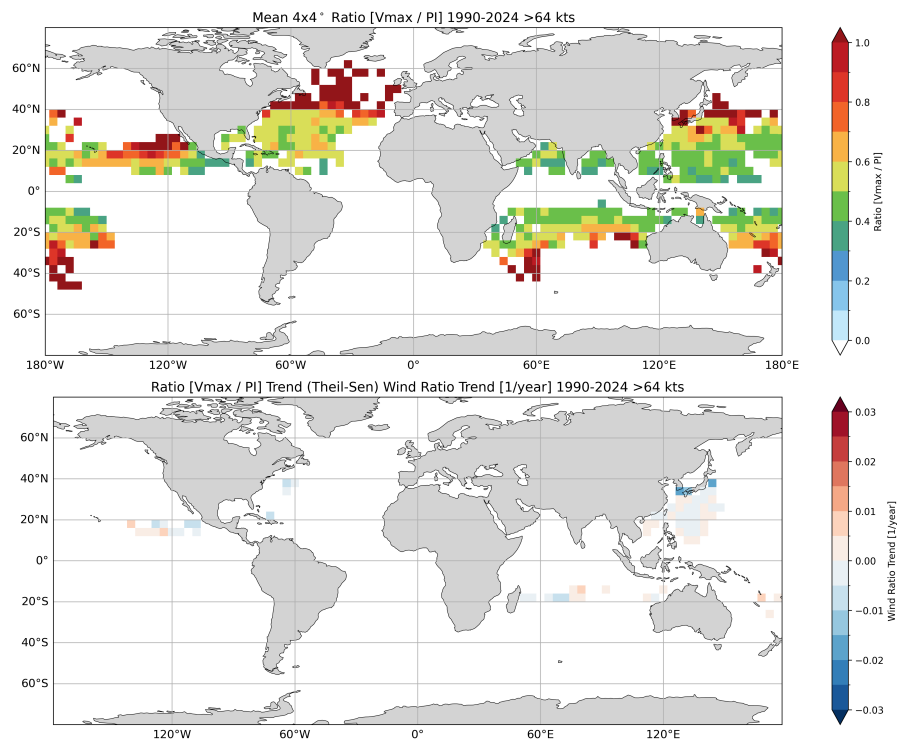


Figure 4: Mean ratio of observed maximum wind speed ( $V_{max}$ ) to potential intensity (PI) and corresponding temporal trends from 1990–2024 for hurricane-strength systems. Values near 1 indicate storms approaching their theoretical maximum intensity.

The ratio trends are very weak to be able to determine whether the strong storms are indeed reaching their intensification. We do see a very weak mix of both positive and negative trends but it is not as significant for a conclusion. This metric fails to answer whether max wind intensity shifts with potential intensity.

### 3.4 Sensitivity Tests

#### 3.4.1 Weak Systems

Other than filtering out just the strong systems, we tested the trend estimation by including weaker systems as well. This would be wind speeds of at least 34 knots or 17 m/s. The trend map for wind intensity differed from what we saw just for the strong systems. Positive trends are sparse and small ( $+0.35 \text{ m/s yr}^{-1}$  to  $+0.70 \text{ m/s yr}^{-1}$ ) in certain regions of North Atlantic and South Indian. In contrast, WNP, parts of NA, SI, and Australia exhibited negative trends ( $-0.70 \text{ m/s yr}^{-1}$  to  $-1.05 \text{ m/s yr}^{-1}$ ). These declining mean intensities in several basins indicates that including weaker systems does not lead to overall strengthening of the TCs but may have slightly weakened in parts of the tropics over the past 40 years.

A trend analysis was ran on the Vmax/PI ratio for the weak systems which revealed very weak trends. Small positive trends appear in the North Atlantic, parts of WNP and eastern South Indian while equally small negative trends appear in the Caribbean, Bay of Bengal, and Madagascar. Unlike the clear positive trends observed for only the strong systems, the broader population of TCs showed no coherent global or basin-scale increase in this efficiency measure between 1990-2024.

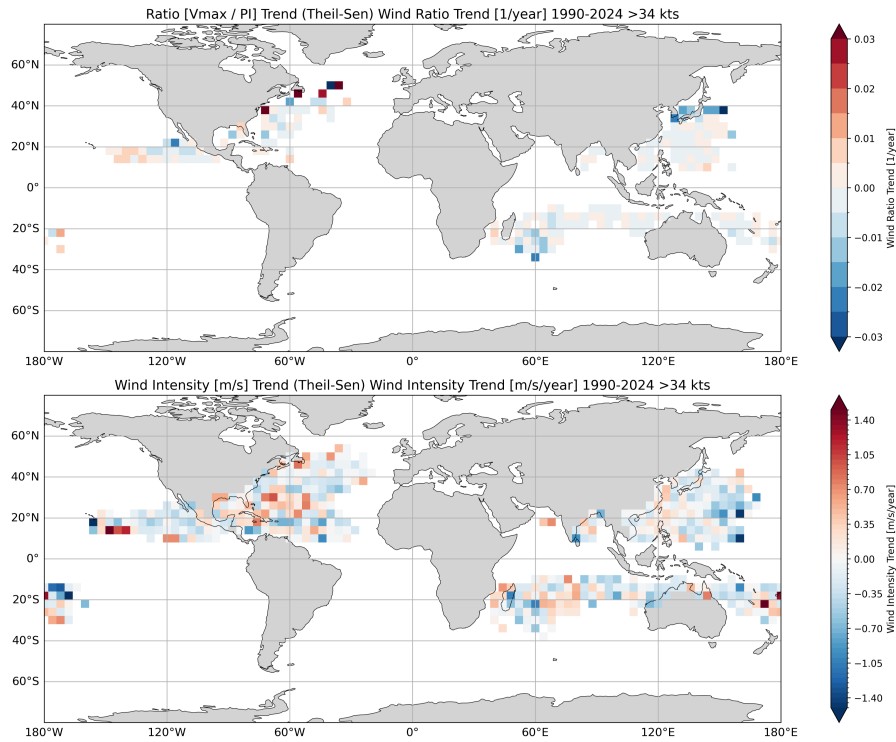


Figure 5: Wind intensity and Vmax/PI trends for a greater range of tropical cyclones, illustrating how inclusion of weaker systems alters the global and basin-scale trend structure relative to strong-storm-only analyses.

These results indicate that determination of intensification using Vmax/PI ratio is difficult. Visually, the results suggests when comparing Figures 4 and 5, we see the ratio is more intense when including the weak systems in the North Atlantic and WNP. However, the ratio is muted when only considering the strong systems. Perhaps certain storms that reached peak intensification were excluded when taking in a smaller subset.

### 3.4.2 Minimum Sea-Level Pressure Trends

Mean minimum sea-level pressure (Pmin, in mb) for strong hurricanes from 1990–2024 indicates the centers of storm intensity, with lower pressures indicating stronger systems. The lowest mean Pmin values (below 950–960 mb) are concentrated in the western North Pacific around the Philippines and Japan, which is consistent with the basin’s history of producing exceptionally low-pressure storms. Similarly low pressures (960–980 mb) is seen in the North Atlantic basin region, including the Caribbean and Gulf of Mexico. Higher mean pressures (approaching 990–1010 mb) dominate less active basins such as the eastern North Pacific, the North Indian Ocean, and portions of the Southern Hemisphere, corresponding to generally weaker cyclones in these regions.

Trends in Pmin, again, revealed localized areas of strengthening and weakening. The results show negative trends as large as  $+1.05 \text{ mb yr}^{-1}$  in parts of the western North Pacific south of Japan and in sections of the South Indian, indicating a possible worsening of hurricane intensification in these region. Negative trends of up to approximately  $-1.40 \text{ mb yr}^{-1}$  appear in the North Atlantic, and region east of Australia which suggests intensification.

**Research Question:** Is there a shift in mean surface level pressure ? With lower central pressure indicating a stronger storm.

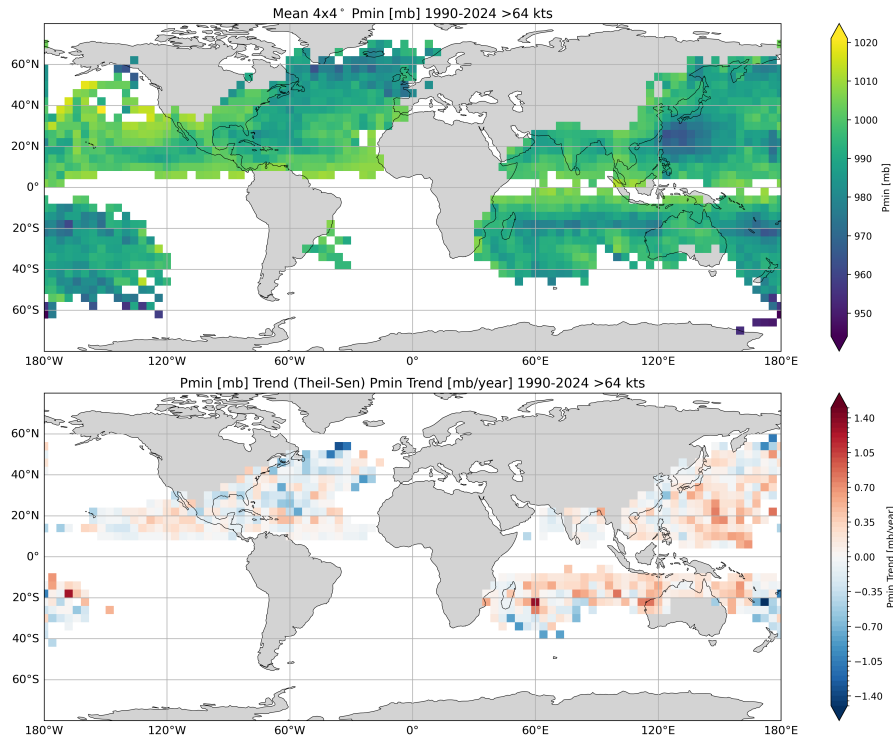


Figure 6: Mean minimum sea-level pressure (Pmin) and linear trend estimates for hurricane-strength tropical cyclones (1990–2024). Negative trends represent long-term pressure deepening and intensification as seen in North Atlantic.

The negative trends in pressure minimum as seen in the North Atlantic and a few regions in the WNP suggest intensification of strong systems in these regions whereas positive trends in majority of the region in the WNP and the south Indian suggest weakening of strong systems. This result checks out with the wind intensity trends in NA (Figure 2) where a similar trend of increasing wind speeds is seen for negative pressure trend. This result is however different for WNP where a positive trend in pressure is seen as well as an increase in overall wind speeds is seen as well.

### 3.4.3 Periodic Tests (All Systems)

To evaluate the robustness of tropical cyclone (TC) intensity trends and potential non-stationarities over time (Figure 7), the IBTrACS dataset was partitioned into multiple temporal segments: the full record (1990–2024), an earlier period (1990–2010), and a later period (2000–2024). Trends were computed for each interval to detect temporal shifts that may be associated with decadal variability or externally forced climate signals.

Global trends are weak and spatially fragmented for **the entire period**. Small positive trends ( $+0.35$  to  $+0.70 \text{ m s}^{-1} \text{ yr}^{-1}$ ) are scattered in the subtropical North Atlantic and parts of the northern WNP, while modest negative trends ( $-0.35$  to  $-0.70 \text{ m s}^{-1} \text{ yr}^{-1}$ ) appear in the Caribbean, Eastern North Pacific, South China Sea, and near Australia. Unlike the clear intensification signal seen in major hurricanes ( $\geq 64 \text{ kt}$ ), the broader tropical cyclone population shows no coherent global or hemispheric strengthening over the 35-year record.

**During the earlier period (1990–2010)**, This interval exhibits the strongest and most widespread negative intensity trends of the three periods. Large areas of the North Atlantic (especially the Caribbean and Gulf of Mexico), Eastern North Pacific, Western North Pacific south of Japan, and the South Indian Ocean show declines of  $-0.70$  to  $-1.40 \text{ m s}^{-1} \text{ yr}^{-1}$ . Positive trends are almost entirely absent.

**The later period (2000–2024)**, negative trends weaken substantially as compared to (1990–2010), while positive trends ( $+0.35$  to  $+1.05 \text{ m s}^{-1} \text{ yr}^{-1}$ ) emerge in the central and northern North Atlantic, northern ENP, and parts of South Indian. Several tropical regions that previously exhibited strong declines (e.g., the Caribbean and Eastern North Pacific) now show near-zero or slightly positive trends. This shift toward intensification in the most recent two decades aligns closely with the rapid global warming and record-high sea surface temperatures observed since  $\sim 2000$ .

**Research Question:** Is this positive shift pertaining to a certain group of Tropical Cyclones or the entirety of it ?

The divergence between earlier decades to recent, shows intensification and demonstrates that this positive shift in TC intensity changes over time and this change is driven primarily by stronger systems and not by the entire population of TCs equally. Another likely cause is possibly advancement in technology as discussed below.

## 3.5 Comparison of HURSAT and IBTrACS

HURSAT provides a homogenized, satellite-based reanalysis of tropical cyclone (TC) intensities designed to minimize biases arising from evolving observational technologies. In contrast, IBTrACS aggregates best-track data from multiple operational meteorological agencies, which may introduce heterogeneous reporting practices across regions and decades.

Spatially, the Theil–Sen trend maps derived from HURSAT do not conclusively any wind intensification trends. Just based on visual evidence, it is observed that there is both

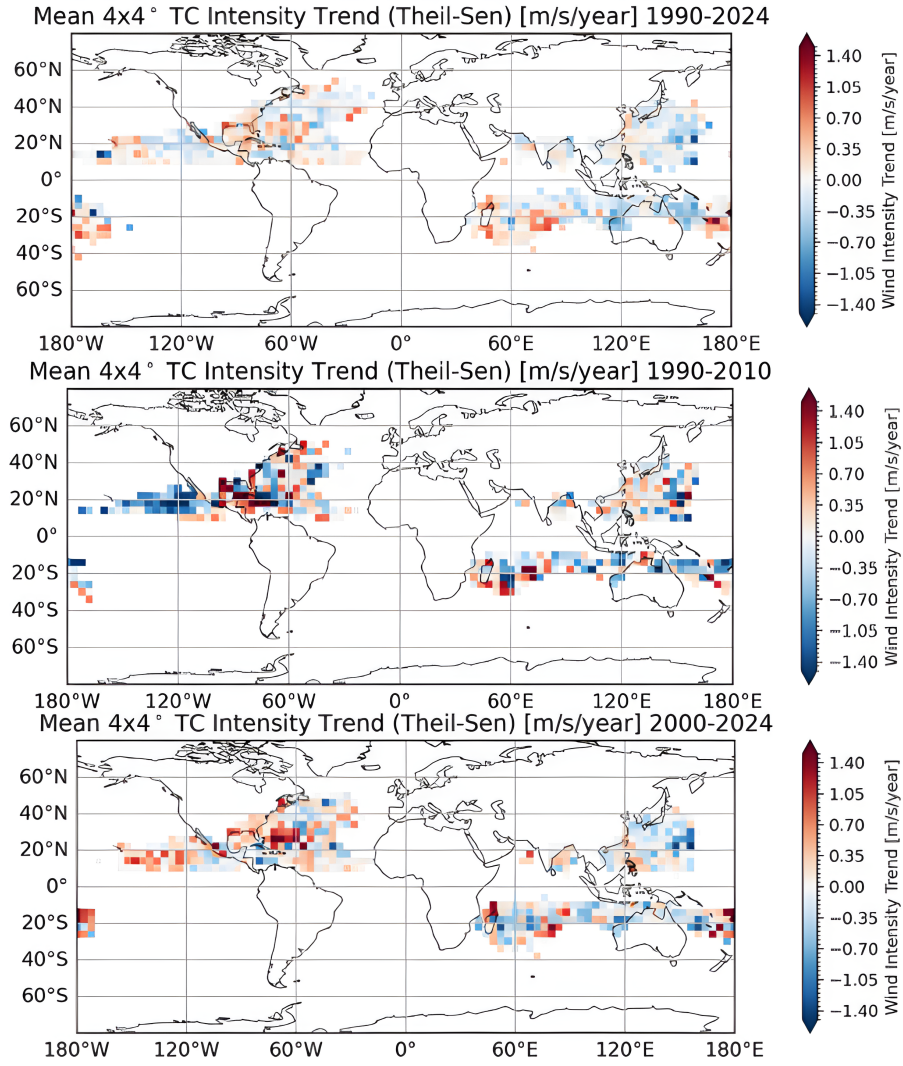


Figure 7: Comparison of tropical cyclone intensity trends for three periods: 1990–2024, 1990–2010, and 2000–2024. These maps highlight non-stationarity in long-term TC behavior and the emergence of intensification in recent decades.



intensification and worsening of between  $+0.35 \text{ m s}^{-1} \text{ yr}^{-1}$  to  $-0.35 \text{ m s}^{-1} \text{ yr}^{-1}$  across all basins.

IBTrACS trend maps reveal broader regions of both positive and negative trends, also reaching  $-1.05 \text{ m s}^{-1} \text{ yr}^{-1}$ , particularly in the Caribbean and Gulf of Mexico within the NA, the WNP near Japan, and the ENP off the coast of Mexico. These signals generally align with those from HURSAT but often cover a larger spatial area and exhibit stronger gradients. Positive trends in IBTrACS (up to  $+0.70 \text{ m s}^{-1} \text{ yr}^{-1}$ ) appear in similar hotspots—including the central NA and northern WNP—but with additional clusters evident in the central North Pacific (CNP) and SI.

Notable discrepancies occur in regions such as Oceania, where IBTrACS displays stronger negative trends in the SP near Australia, whereas HURSAT shows more neutral or weakly positive signals. Despite these differences, both datasets sort of agree on some displacement of positive trends, possibly reflecting climate-driven shifts in TC genesis or intensification latitudes. The greater variability in IBTrACS trends may stem from agency-specific analysis techniques or evolving best-track standards.

**Research Question:** Why do we observe differences in tropical cyclone intensities when comparing datasets that use different observational techniques?

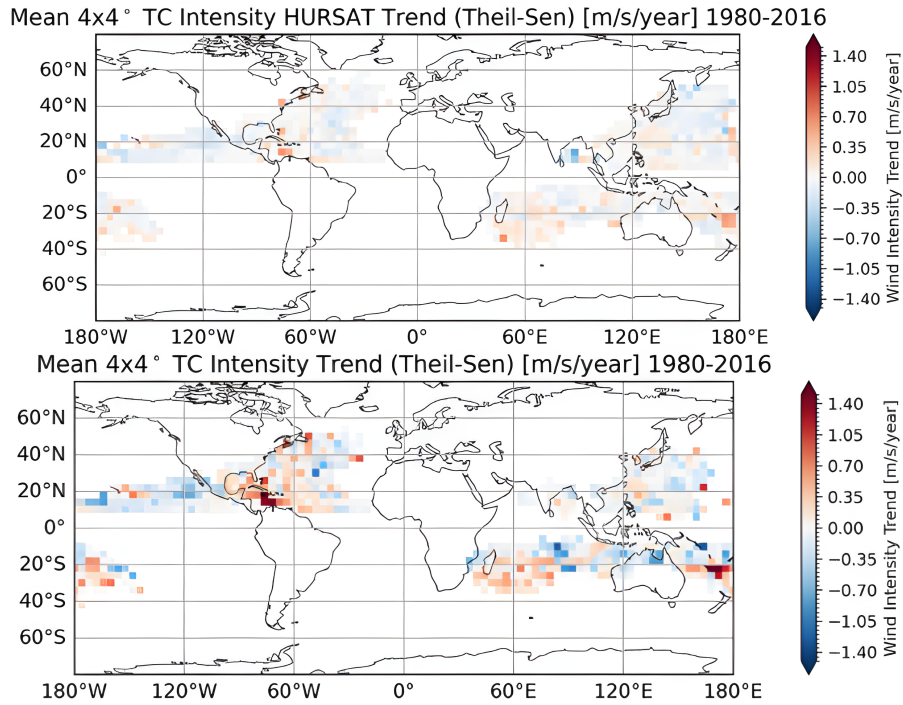


Figure 8: Comparison of wind intensity trends using the homogenized HURSAT dataset and the IBTrACS best-track dataset. Differences illustrate the influence of observational methods and measurement heterogeneity on trend estimation. HURSAT is visually muted as compared to IBTrACS.

### 3.5.1 Basin-Specific Comparison

**North Atlantic:** Both datasets show negative trends from  $0^{\circ}$ – $20^{\circ}$ N (approximately  $-2$  to  $-4 \text{ m s}^{-1} \text{ decade}^{-1}$ ), transitioning to near-zero or slightly positive values at mid-latitudes ( $30^{\circ}$ – $50^{\circ}$ N). IBTrACS exhibits stronger variability and deeper negative values near  $40^{\circ}$ N.

**Western North Pacific:** IBTrACS shows sharper fluctuations, with negative trends down to  $-10 \text{ m s}^{-1} \text{ decade}^{-1}$  between  $40^\circ$  and  $50^\circ\text{N}$  and positive trends around  $15^\circ\text{N}$ . The trends results are significantly different for both datasets beyond  $40^\circ\text{N}$ .

**Central North Pacific:** Both datasets display consistent negative trends across  $0^\circ$ – $40^\circ\text{N}$  (approximately  $-2$  to  $-4 \text{ m s}^{-1} \text{ decade}^{-1}$ ), indicating robust weakening signals in this basin. IBTrACS has more variability as compared to HURSAT.

**Eastern North Pacific:** IBTrACS trends show more pronounced declines ( $-2$  to  $-3 \text{ m s}^{-1} \text{ decade}^{-1}$ ) between  $15^\circ$  and  $20^\circ\text{N}$ . Both dataset follow a similar pattern for up and down trend at each latitude with IBTrACS, having a broader trend variability.

**Southern Indian:** Patterns are more variable, with IBTrACS showing positive trends at  $20^\circ$ – $30^\circ\text{S}$  (up to  $+4 \text{ m s}^{-1} \text{ decade}^{-1}$ ) and negative trends closer to the equator. HURSAT tends to have greater positive trend as compared to IBTrACS in this basin.

**South Pacific:** Both datasets indicate similar trends from  $40^\circ\text{S}$  to the equator ( $-3$  to  $+6 \text{ m s}^{-1} \text{ decade}^{-1}$ ) to  $20^\circ\text{S}$ , though IBTrACS shows larger amplitudes. This trend however breaks between  $10^\circ\text{S}$  to  $20^\circ\text{S}$  where HURSAT has more positive trends above 0 whereas IBTrACS has more negative trends below 0.

**Global:** Globally aggregated trends show widespread weakening, with IBTrACS values consistently lower than HURSAT within the deep tropics ( $20^\circ\text{S}$ – $20^\circ\text{N}$ ) and convergence at higher latitudes. This suggests that HURSAT reduces upward intensity biases present in raw best-track observations, providing a more homogenized global perspective.

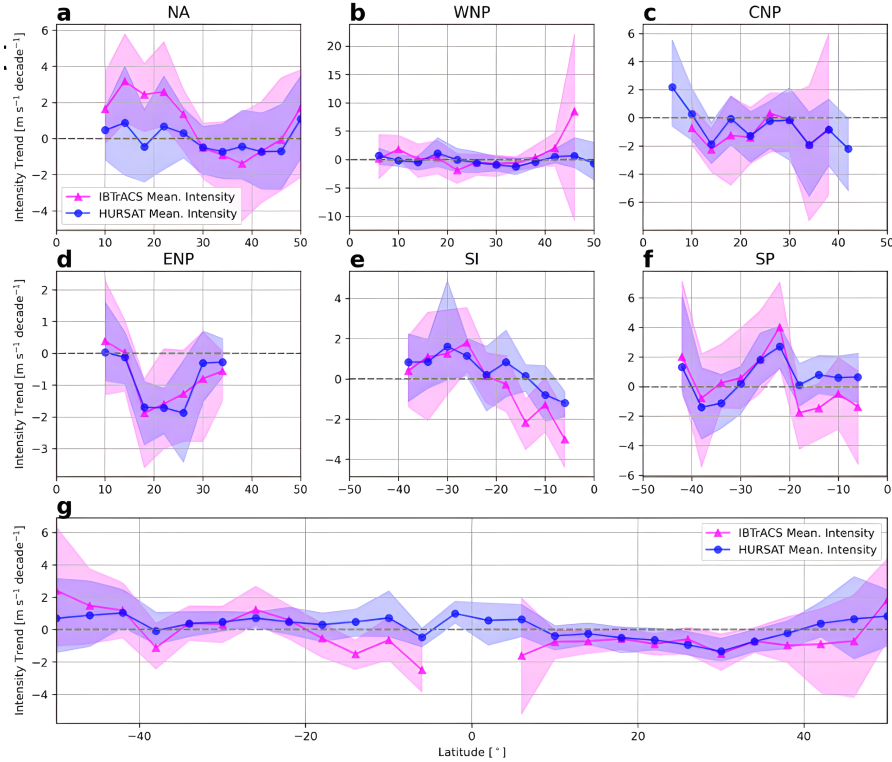


Figure 9: Latitude-intensity trend profiles for each basin comparing IBTrACS and HURSAT. These reveal regional consistencies and discrepancies arising from differing observational methods.

Based on all the observations visually made from the HURSAT v IBTrACS plots, we know that methodological differences in the observation techniques does affect interpretation of trends.

## 4 Conclusion

This project examined historical trends in tropical cyclone intensity using the IBTrACS and HURSAT datasets, focusing on maximum sustained wind speed, minimum sea-level pressure, track density, and the ratio of observed intensity to potential intensity. The analysis reveals that changes in tropical cyclone intensity over 1990–2024 are not globally uniform, but instead vary strongly by basin, storm intensity class, and time period.

Major hurricanes ( $V_{\max} \geq 64$  kt) show the clearest signals of intensification, particularly in the North Atlantic (NA) and "subtly" other basins. Zonal-mean trends confirm that these increases are concentrated in the Northern Hemisphere subtropics in the North Atlantic basin, while other basins such as the eastern North Pacific (ENP), West North Pacific (WNP), and Southern Hemisphere such as the South Indian basin exhibit weaker or mixed trends. **When weaker systems are included, intensification largely disappear, suggesting that recent strengthening is primarily driven by the upper tail of the intensity distribution rather than the entire tropical cyclone population.**

Periodic Test highlighted that intensification has become more apparent in the most recent two decades, consistent with the influence of rapid ocean warming in recent years. Natural climate variability also plays a role: **repeated La Niña episodes during the study period likely contributed to enhanced North Atlantic activity** by reducing vertical wind shear and increasing thermodynamic favorability, while producing the opposite effect in portions of the ENP and WNP. Thus, observed trends reflect the combined influence of long-term warming and shorter-term modes of variability.

Comparison between IBTrACS and the homogenized HURSAT dataset shows broad agreement in the location of intensification and weakening regions, but differences in magnitude due to methodological and observational inconsistencies across agencies. These differences underscore the importance of using multiple datasets when evaluating long-term changes.

Overall, the results support the evidence that the **strongest** tropical cyclones have intensified in the modern warming climate induced by anthropogenic activities.

## References

- Kossin, J. P., Knapp, K. R., Olander, T. L., & Velden, C. S. (2020). *Global increase in major tropical cyclone exceedance probability over the past four decades*. Retrieved 2025-11-11, from <https://repository.library.noaa.gov> (ISSN: 22 Volume: 117) doi: 10.1073/pnas.1920849117
- Kruskie, A., & Chavas, D. (n.d.). Interbasin Analysis of the Poleward Expansion of Tropical Cyclone Potential Intensity. Retrieved 2025-11-13, from <https://www.authorea.com/users/767230/articles/1306731-interbasin-analysis-of-the-poleward-expansion-of-tropical-cyclone-potential-intensity>